De La Salle Holy Cross College

NOVEMBER EXAMINATIONS

2024

Grade 10

FURTHER STUDIES MATHEMATICS

Probability, Statistics and Algebra

Examiner:	Ms A. Bux	**Time:**	2 hours
Moderator:	Mr J. Blockley	**Marks:**	105

Name: Marking Guidelines Final

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 9 questions and 20 pages that include 3 pages for extra calculations or rough work.
2. Please check that your question paper is complete.
3. Answer the questions in the space provided on this question paper.
4. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated. Ensure that it is set in DEGREE mode.
5. ALL necessary calculations must be shown clearly. Answers only will not necessarily be awarded full marks.
6. Diagrams and graphs are not necessarily drawn to scale.
7. Round off final answers to **THREE DECIMAL PLACES**, unless stated otherwise.
8. It is in your own interest to write legibly and to present your work neatly.
9. Three blank pages (pages 11, 19 and 20) are included in the paper. If you run out of space for a question, then use this page. Clearly indicate the question number of your answer should you use this extra space.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Total
Mark										
Total	10	15	11	14	10	10	10	12	13	105

Formulae you may need:

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

SECTION A – ALGEBRA

QUESTION 1

State whether the following statements are TRUE or FALSE. If FALSE, correct the statement.

(a) The imaginary unit is defined as $i = \sqrt{-1}$ and therefore $i^2 = -1$. (2)

True ✓ a

(b) The absolute value function $f(x) = |x|$ has a minimum value of zero. (2)

True ✓ a

(c) The equation $|-x| = -x$ is true for all real values of x. (2)

False ✓, $|-x| = -x$ for $x < 0$ ✓ or $|-x| = x$

(d) $|x - 3| = 3 - x$ for all real values of x. (2)

False, $|x-3| = |3-x|$ or

$|x-3| = 3-x$ when $x \leq 3$

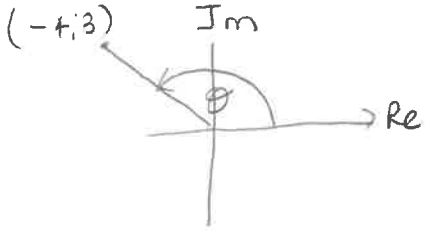
or

$|x-3| = x-3$ when $x \geq 3$

(e) The polar coordinates of $-4 + 3i$ is $(5;53,1°)$. (2)

False

$(5; 143,130^\circ)$



$r = 5$

$\theta = 180 - \tan^{-1}\left(\frac{3}{4}\right)$

$= 143,130^\circ$

don't penalise for dp

[10]

QUESTION 2

(a) Simplify fully:

$$\frac{1}{x^2-4x+4}+\frac{1}{x^2-6x+9}-\frac{1}{x^2-5x+6} \qquad (6)$$

$$= \frac{1}{(x-2)^2} + \frac{1}{(x-3)^2} - \frac{1}{(x-2)(x-3)}$$

$$= \frac{(x-3)^2 + (x-2)^2 - 1(x-2)(x-3)}{(x-2)^2(x-3)^2} \qquad \text{LCD} = (x-2)^2(x-3)^2$$

$$= \frac{x^2-6x+9 + x^2-4x+4 - (x^2-5x+6)}{(x-2)^2(x-3)^2}$$

$$= \frac{2x^2-10x+13-x^2+5x-6}{(x-2)^2(x-3)^2}$$

$$= \frac{x^2-5x+7}{(x-2)^2(x-3)^2}$$ ca

(b) Determine the value of:

$$1+\cfrac{6}{1+\cfrac{6}{1+\cfrac{6}{1+\ldots}}} \qquad (5)$$

Let $1+\cfrac{6}{1+\cfrac{6}{1+\ldots}} = x$

$$1+\frac{6}{x} = x$$

$$x^2-x-6 = 0$$

$$(x-3)(x+2) = 0$$

$x = 3$ or $x \neq -2$

(c) Show that the roots of $x^2 + 2ax + 3a^2 = 0$ are non-real for all real values of a, and $a \neq 0$. (4)

Should have stated without solving the equation.

$x^2 + 2ax + 3a^2 = 0$

$b^2 - 4ac = (2a)^2 - 4(1)(3a^2)$ ✓

$= 4a^2 - 12a^2$

$= -8a^2$ ✓

$a^2 > 0$ ✓ $\therefore -8a^2 < 0$ ✓ } There must be an explanation of why $\Delta < 0$! [15]

Since $a \neq 0$

$\Rightarrow$ Roots are non-real

Solving the eqn:

$x = \dfrac{-2a \pm \sqrt{(2a)^2 - 4(1)(3a^2)}}{2(1)}$ ✓

$= \dfrac{-2a \pm \sqrt{4a^2 - 12a^2}}{2}$

$= \dfrac{-2a \pm \sqrt{-8a^2}}{2}$ ✓

$= \dfrac{-2a \pm a2\sqrt{2}\,i}{2}$

$= -a \pm a\sqrt{2}\,i$ ✓

$\Rightarrow$ Roots are Non-real.

QUESTION 3

Solve for $x \in \mathbb{C}$, where necessary. Leave answers in simplified surd form where necessary.

(a) $|2x - 5| = 6 - 3x$ (6)

$2x-5 = 6-3x$ ✓ if $2x-5 \geq 0$ ✓

$5x = 11$

$x \neq 11/5$ ✓

$2x \geq 5$

$x \geq 5/2$

$2x-5 = -6+3x$ ✓ if $2x-5 < 0$ ✓

$-x = -1$

$x = 1$ ✓

$2x < 5$

$x < 5/2$

Valid

or must check / eliminate non-solution for full marks.

(b) $x^3 + x(3x + 5) + 3 = 0$ (5)

$x^3 + 3x^2 + 5x + 3 = 0$

$(-1)^3 + 3(-1)^2 + 5(-1) + 3 = -1 + 3 - 5 + 3 = 0$

$\therefore (x+1)$ is a factor of $x^3 + 3x^2 + 5x + 3 = 0$

$x^3 + 3x^2 + 5x + 3 = 0$

$(x+1)(x^2 + bx + 3) = 0$ ✓

$(x+1)(x^2 + 2x + 3) = 0$ ✓

$bx^2 + x^2 = 3x^2$

$bx^2 = 2x^2$

$\therefore b = 2$

$x = -1$ ✓ or $x = \dfrac{-2 \pm \sqrt{(2)^2 - 4(1)(3)}}{2(1)}$ ✓ Sub

$= \dfrac{-2 \pm \sqrt{4-12}}{2}$

$= \dfrac{-2 \pm \sqrt{-8}}{2}$

$= -1 + \sqrt{2}i$ or $-1 - \sqrt{2}i$ ✓ both

1,414 or -1,414

$x+1 \;\big|\; x^3 + 3x^2 + 5x + 3$

$x^2 + 2x + 3$

$x^3 + x^2$

$2x^2 + 5x$

$2x^2 + 2x$

$3x + 3$

$3x + 3$

0

R

[11]

QUESTION 4

(a) Express the following in the form $a + bi$. (4)

$$\frac{3+i}{3-i} \times \frac{3+i}{3+i}$$ ✓ × conj

$$= \frac{9+6i+i^2}{9-i^2}$$ ✓

$$= \frac{8+6i}{10}$$ ✓

$$= \frac{4}{5} + \frac{3}{5}i$$ ✓ form $a+bi$

(b) Determine the value(s) of p and t in the equation: (6)

$$(3i+5)(ti+p) = -2i+1$$

$$ti+p = \frac{-2i+1}{3i+5} \times \frac{3i-5}{3i-5}$$ ✓ × conj

$$= \frac{-6i^2+10i+3i-5}{9i^2-25}$$ ✓

$$= \frac{1+13i}{-34}$$ ✓

$$= -\frac{1}{34} - \frac{13}{34}i$$ ✓

$$t = -\frac{13}{34} \quad ; \quad p = -\frac{1}{34}$$ ✓ ✓

OR

$$3ti^2 + 3pi + 5ti + 5p = -2i + 1$$ ✓ ✓

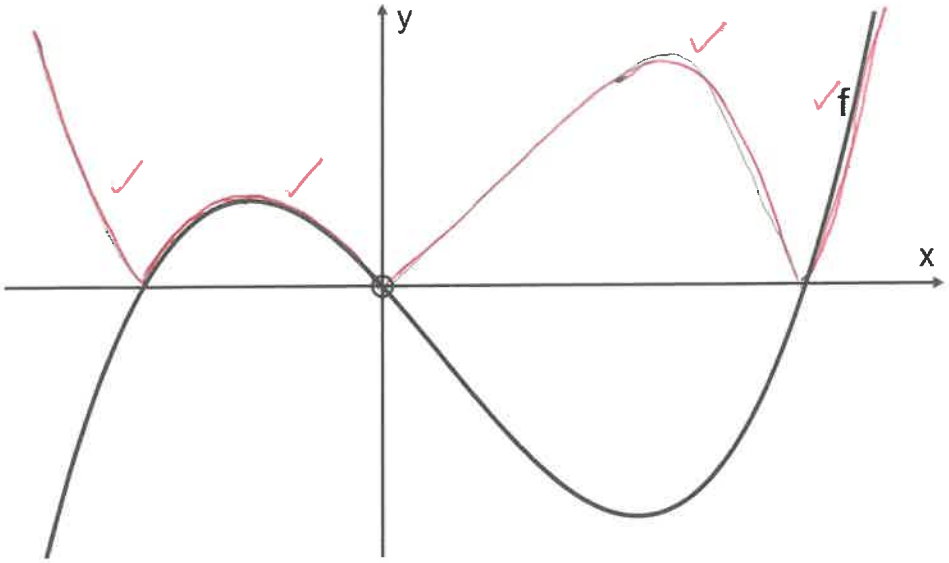
$$-3t + 5p + (5t+3p)i = 1 - 2i$$

$$-3t + 5p = 1 \quad \times 5$$
$$5t + 3p = -2 \quad \times 3$$ ✓

$$-15t + 25p = 5$$
$$15t + 9p = -6$$ ✓
$$34p = -1$$
$$p = -\frac{1}{34}$$ ✓

$$5t + 3\left(-\frac{1}{34}\right) = -2$$
$$t = -\frac{13}{34}$$ ✓

(c) Sketched below is the graph of $f(x) = x^3 - x^2 - 3x$.

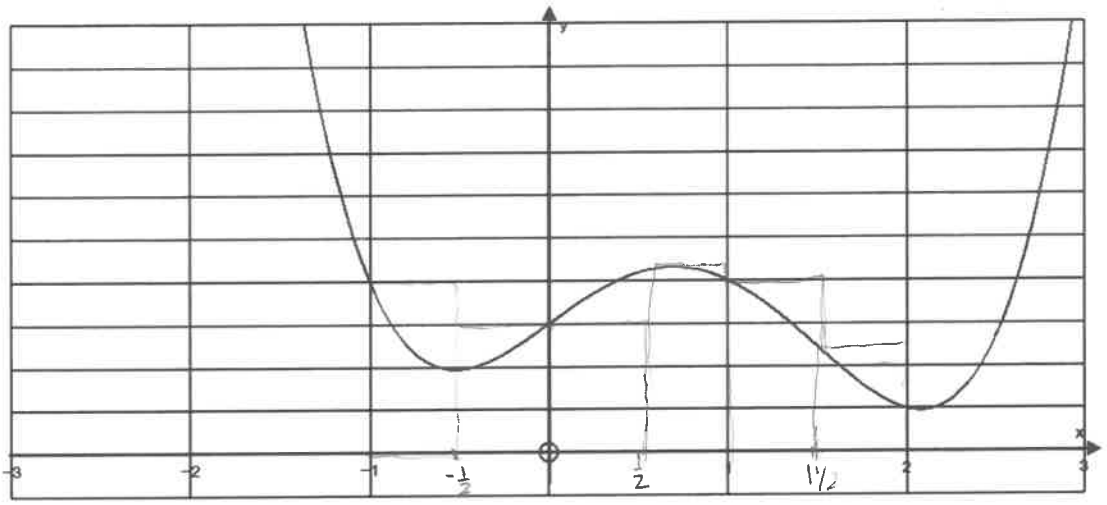


$g(x) = |f(x)|$ Using a different colour pencil, sketch $g(x)$ on the axes provided above. (4)

[14]

QUESTION 5

(a) Using left endpoint approximation, estimate the area between the graph of the function $f(x) = x^4 - 3x^3 + 3x + 3$ and the x-axis using equally spaced partitions: $\left\{-1; -\frac{1}{2}; 0; \frac{1}{2}; 1; \frac{3}{2}; 2\right\}$. (5)



Estimated Sum $= \left[f(-1) \times \frac{1}{2}\right] + \left[f(-\frac{1}{2}) \times \frac{1}{2}\right] + \left[f(0) \times \frac{1}{2}\right] + \left[f(\frac{1}{2}) \times \frac{1}{2}\right] + \left[f(1) \times \frac{1}{2}\right] + \left[f(\frac{3}{2}) \times \frac{1}{2}\right]$ ✓

$= \frac{1}{2}\left[4 + \frac{31}{16} + 3 + \frac{67}{16} + 4 + \frac{39}{16}\right]$ ✓m ✓ma

$= \frac{1}{2}\left[\frac{313}{16}\right]$

$= \frac{313}{32}$

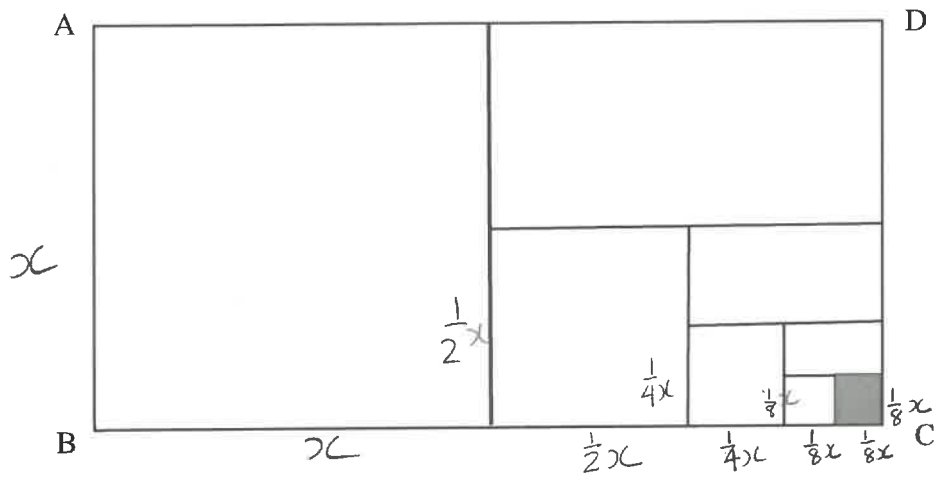
$\approx 9,781$ units^2 ✓ca

$\frac{31}{16} = 1,9375$

$\frac{67}{16} = 4,1875$

$\frac{39}{16} = 2,4375$

(b)



In the rectangle above, BC = 2AB, and BC has been repeatedly divided in half resulting in the figure.
What fraction of the area of rectangle ABCD is the area of the small shaded square? (5)

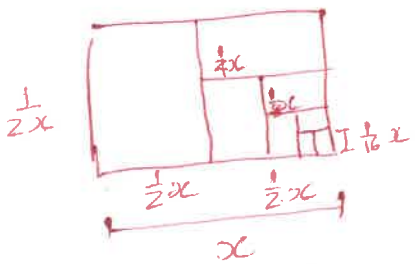
Area ABCD = $2x(x) = 2x^2$ ✓

Area shaded square = $\frac{1}{8}x \times \frac{1}{8}x$ ✓

$= \frac{x^2}{64}$ ✓

$\frac{\frac{x^2}{64}}{2x^2}$ ✓M $= \frac{x^2}{64} \times \frac{1}{2x^2} = \frac{1}{128}$ ✓CA1 **[10]**

OR



$\frac{\frac{1}{16}x \times \frac{1}{16}x}{\frac{1}{2}x \times 2x}$

$= \frac{\frac{x^2}{256}}{\frac{x^2}{2}}$

$= \frac{1}{128}$

ADDITIONAL SPACE (ALL QUESTIONS)

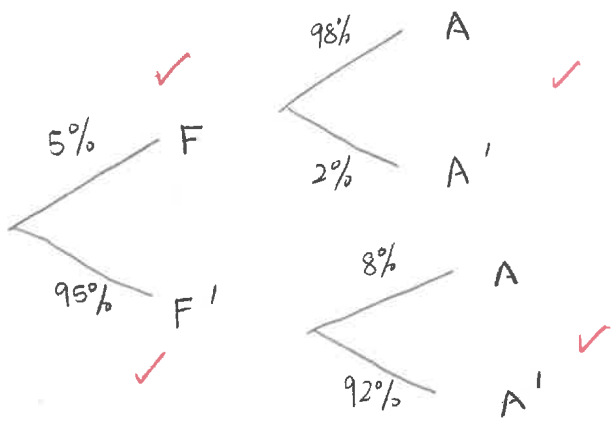
REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL QUESTIONS ARE MARKED.

SECTION B – PROBABILITY

QUESTION 6

An airport screens bags for forbidden items, and an alarm is supposed to be triggered when a forbidden item is detected. If a bag contains a forbidden item, there is a 98% chance that it triggers the alarm. If a bag does not contain a forbidden item, there is an 8% chance that the alarm is triggered. Suppose that there is a 5% chance that a bag contains a forbidden item.

(a) Draw a tree diagram to represent the given information. (4)



(b) Determine the probability that a randomly selected bag triggers the alarm. (3)

$(5\% \times 98\%) + (95\% \times 8\%)$

$= \frac{1}{8}$ or $0,125$ ✓ CA

(c) Given that the alarm triggers, what is the probability that the bag contains a forbidden item? (3)

$P(F/A) = \frac{P(F \cap A)}{P(A)}$

$= \frac{5\% \times 98\%}{0,125}$

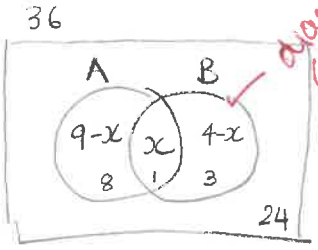
$= \frac{49}{125}$ or $0,392$ ✓ CA

[10]

QUESTION 7

A and B are two events in sample space S where $n(S) = 36$; $n(A) = 9$; $n(B) = 4$ and $n(\text{not}(A \text{ or } B)) = 24$.

Determine:



(a) $P(A \text{ or } B)$ (3)

$n(A \text{ or } B) = 36 - 24 = 12$

$P(A \text{ or } B) = \frac{12}{36} = \frac{1}{3}$

$9 - x + x + 4 - x + 24 = 36$

$1 = x$

$\therefore P(A \text{ or } B) = \frac{12}{36}$

(b) $P(A \text{ and } B)$ (4)

$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$ from

$\frac{1}{3} = \frac{9}{36} - \frac{4}{36} - P(A \cap B)$ from

$P(A \cap B) = \frac{1}{36}$

or from venn diagram above or $\frac{1}{36}$

(c) Whether events A and B are independent. Justify your answer with calculations. (3)

$P(A) = \frac{9}{36}$ $P(B) = \frac{4}{36}$

$P(A) \times P(B) = \frac{9}{36} \times \frac{4}{36} = \frac{1}{36}$ $P(A \cap B) = \frac{1}{36}$

Since $P(A \cap B) = P(A) \times P(B)$

[10]

Events A and B are independent events

QUESTION 8

(a) How many codes of the form *XYYYX* can be made where *X* represents any letter of the alphabet and *Y* represents any digit from 0 to 5? The letters may not be repeated, but the numbers may be repeated. (4)

26 6 6 6 25

$26 \times 6^3 \times 25$

= 140 400 ca

(b) In a class, five pupils need to be given their seating positions, on five chairs, from the front of the class to the back.

(1) In how many ways can the pupils be organized, on the five chairs, from the front to the back? (2)

$5! = 120$

(2) In how many different ways can the pupils be seated, if two of the pupils, Naledi and Vincent, need to be seated in the front two chairs? (3)

N A _ _ _

$3! \times 2!$

= 12 ca

(3) In how many different ways can the pupils be organized if there are three girls and two boys, and the girls and boys have to alternate? (3)

G B G B G

$3! \checkmark \quad 2! \checkmark$

$= 12 \checkmark$

[12]

QUESTION 9

A company has 1825 employees. The employees are classified as professional, skilled or elementary.
The following table shows:

- the number of employees in each classification
- the two areas, A or B, where the employees live

	A	B	
Professional	740	380	1120
Skilled	275	90	
Elementary	260	80	340
	1275		

An employee is chosen at random.

Determine the probability that this employee:

(a) is skilled. (2)

$\frac{275+90}{1825} = \frac{365}{1825} = \frac{1}{5}$ or 0,2

(b) lives in area B and is not professional. (2)

$\frac{90+80}{1825} = \frac{170}{1825} = \frac{34}{365}$ or 0,093

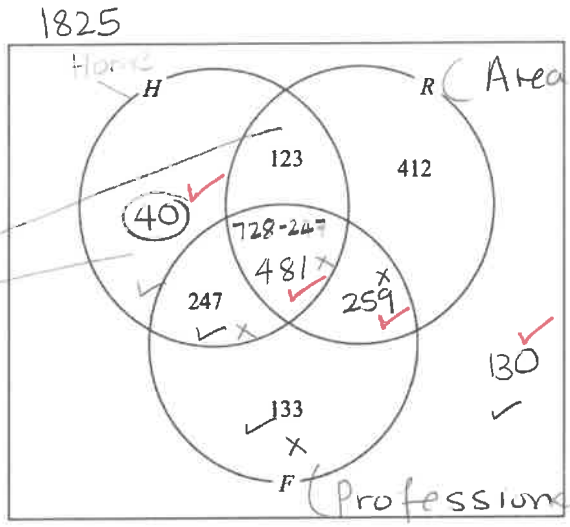
(c) Some classifications of employees are more likely to work from home.

- 65% of **professional** employees work from home
- 40% of skilled employees work from home
- 5% of elementary employees work from home
- Event F is that the employee is a professional
- Event H is that the employee works from home
- Event R is that the employee is from area A

65% × 1120 = 728

40%(275 + 260) + 5%(340)

(1) Using the information given above, complete the following Venn diagram: (4)



40%(535) + 5%(340)

= 214 + 17

= 163

163 − 123

= 40

(2) Hence, determine:

(i) $P(R' \cap F)$ (2)

$$\frac{133+247}{1825} = \frac{380}{1825}$$

Penalise once if answer not as probability

(ii) $P([H \cup R]')$ $= \frac{133+130}{1825} = \frac{263}{1825}$ (1)

iii) $P(F/H) = \frac{247+481}{40+123+481+247}$

$= \frac{728}{891}$ (2)

(iii) $P(F|H)$ $= \frac{247+481}{1825}$ (2)

$\frac{247+481+123+40}{1825}$

$= \frac{728}{891}$ ✓ca ✓ca

[13]

or 0,817

105 marks

or

$\frac{P(F \cap H)}{P(H)} = \frac{247+481}{891}$

ADDITIONAL SPACE (ALL QUESTIONS)

REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE ADDITIONAL SPACE TO ENSURE THAT ALL QUESTIONS ARE MARKED.

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